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BILLIARD LIGHT

Abstract

A lighting system for a billiard table comprising a frame, a plurality of lights disposed in the frame, and at least one lens coupled to the frame. The at least one lens is configured to focus a light produced by at least one light disposed in the frame.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is continuation application of Ser. No. 17/900,794 filed on Aug. 31, 2022, now issued as U.S. Pat. No. 12,311,253 which issued on May 27, 2025, which is a non-provisional application which hereby claims priority to U.S. Provisional Application Ser. No. 63/239,357 filed on Aug. 31, 2021, the disclosure of all of these applications are hereby incorporated herein by reference in their entirety.

BACKGROUND OF THE INVENTION

[0002] Generally, when lighting a billiard table, the state of the art has been limited either to illuminate with a centrally hung chandelier of some type which typically casts a disproportionate concentrated illumination toward the middle of the playing surface or blankets the entire area above the table with an overly diffused light in an attempt to create an environment of even lighting across the playing surface. In practice, neither of these approaches is optimal for filming, playing or training on a billiard table.

[0003] The billiard table surface, unlike a conference room table, has two distinct depths. The table bed sits below the nose of the rail cushion by 35-37 mm. To improve playability and enhance the quality of any presentation of the game, the nose of the cushion is an important feature of the table that needs to be carefully and specifically lit in order to be clearly visible from a wide range of perspectives.

[0004] Employing mirrors and diffusers within light sources in billiard table lighting, scatters rays laterally which softens the shadows and floods light under the rail, which creates a situation where it is difficult to visually discern the “nose” of the cushion against the backdrop of the table bed, because both surfaces become lit equally brightly and both surfaces are covered by the same fabric. Lit with this approach, it becomes extremely difficult to discern the precise location of the rubber “nose” of the cushion against the backdrop of the table bed from a majority of overhead perspectives. When the depth of the cushion boundary is not rendered clearly on television, it is difficult for the fans to follow the action, and the production suffers greatly. In order to solve this problem, a completely opposite objective is required. Instead, there is a need for a lighting system which creates a hard shadow between the nose of the cushion, all around the table perimeter, to contrast its depth against the playing surface behind it. Using mirrors to redirect rays horizontally or employing diffusers designed to scatter the light, does exactly the opposite of what is necessary to achieve which is a hard light line providing a stark demarcation of the nose of the cushion.

SUMMARY OF THE INVENTION

[0005] One embodiment of the invention relates to an overhead lighting platform designed to facilitate precise illumination and provide for the aligned mounting of and control of precise LED and laser projections on and around a billiard playing arena. The target projection area is not limited to the top playing surface of the table, but also includes the sides of the table, rails, cushions, and the surrounding flooring in the playing space.

[0006] The platform described has precision alignment capability as well as the ability to fine-tune the position and alignment of all mounted components, including perimeter lighting sources and centrally directed light emitting devices (laser points/lines and/or LED projectors, as well as strategically placed sensors to capture light, sound, temperature and humidity. The mounting rig, called a “spine”, uses an integrated laser line to perfectly align and calibrate its base position. Lasers and any other devices that require precision-aligned mounting to be easily centered over the table and fixed into place in a variety of useful and novel applications.

[0007] The downward focused light rays directed by the lenses create a hard shadow directly under it, that renders the nose of the cushion very clearly against the table bed. The lens helps to control the spread of light to create an even distribution that falls off relatively narrowly and certainly

without flooding light under the rail cushion on the opposite side of the one being directly lit from above. Because of the fact that the adjacent light segments work together to concentrate the light more strongly into the corners. Depending on the height that the fixture is hung above the table, there is a necessity to block different amounts of light, in a controlled fashion, from regions along the various lengths of the perimeter so that, overall, while light rays remain focused to create the hard shadow needed directly below, their output is optimized to reduce hot spots. The exact pattern of the light blocking filter all around the perimeter of the fixture, depends on the spread of the lens being used and the height that the fixture is to the table. If the focused lens concentrates too much light into a particular region, the filter is moderated to balance the undesirable contrast. Controlling the focus angle and intensity of the beams is the objective which is very different from a diffuser which spreads light around carelessly which would only serve to diminish the visibly reduce the definition of the subject cushion on the cushion across from the segment directly below the overhead lens. Lit in this way, the billiard table playability and production quality is improved. Thus, filters and lenses are used to sharpen the light instead of diffusers which soften the light.

[0008] When filming billiards for live television or streaming production, it is not only necessary to illuminate the playing surface of the table, but also important to control exactly how lighting is projected onto the entire stage—including where the athletes walk around and shoot from. The platform and rig allows for LED sources and devices to be mounted, not only inside the bounds of the frame, but also extended outwards (via telescoping extensions to the “spine” and “ribs”) and outside of it as well. In addition, inward and outward-facing scoreboard displays are mountable to enable an immersive and interactive gaming experience for the users and fans.

[0009] The overhead platform described is flexible enough to handle a wide range of use cases where extremely precise positioning of overhead projections are needed. In a billiard application, exactly placed points, lines, and images using devices/sources including (laser diodes, lines, or LED projectors) are useful and desirable to have but extremely time-consuming and difficult to deploy, align and calibrate. For any billiard game, it is critical to mark the playing surface with specific key points and lines on the surfaces of the table to support a variety of games played as well as for training purposes. A host of applications on a billiard table would benefit by augmenting the table to support the attachment of precisely targeted overhead light sources. LED laser projectors (or 3D laser ray scan projection devices) can be used to project still or animated images of event logos, sponsors, gameplay indicators or scoring information, can also be projected onto the table surface and/or onto the floor area surrounding the table, pre or post-game, or presented during half-time or commercial break, for example [0010] to be seen by fans and players.

[0011] For training purposes, carefully aligned projectors can be calibrated to focus light (images) directly on top of specific billiard table surfaces such as the rails, cushions to measure bank shots and kick shots for training and visualization purposes. Coupled with interactive software, extremely effective remote coaching can be facilitated. Projections can be static or dynamically generated and animated to support visualizations of ball positions and ball paths over time.

[0012] The projections help to make possible a number of interactive and digitally enhanced experiences on a billiard table. One application and implementation includes virtual gameplay with a remote player where the motions of the balls at one venue are recorded, forwarded, and projected remotely onto a table surface at their opponent's remote location—and visa versa—so that the two opponents can more directly visualize each other's plays and responses and thus simulate the experience of actually playing them in person by watching their opponents remote actions unfold in front of them as projections atop their local equipment.

Another application includes remote (or local) instruction by a coaching where helpful and instructive lines are drawn onto a tablet device by a mentor, which is rendered directly onto a remote student's table to convey the correct guidance to be studied. Such direct guides would drastically augment the real experience of playing billiards while facilitating accelerated learning.

[0013] When playing billiards, there are always a number of important markings permanently

installed on the table (generally, either adhered with a pressure sensitive sticker, inked-in with a pen, or written-in, less permanently, using a fabric pencil, for example.) Table markings are customary and often specific to certain games commonly played on a billiards table. For example, for all games currently played, the “foot spot” is a point on the “foot” side (racking side) of the table to indicate where one should rack the balls. The “head string” is the horizontal line that spans the width of the table which usually designates where the player should break from (“behind the line”). In 14.1 straight pool, the outer boundary of the rack outline is etched onto the table so that players can plan ahead to prepare the positioning of their last ball and continue their continuous runs.

[0014] In carom billiards, certain games require specific “balk lines” to be present on the table in order to regulate legal zones during plays in the game. A set of 5 to 9 spot markings are commonly used to designate where to position balls at the start of the game or during certain moments when balls need to be re-spotted.

[0015] Temporarily drawn markings and lines can be extremely useful training and practice tools. Usually, such markings never get drawn permanently on publicly used tables and are only seen on home-tables or in training centers.

[0016] The downside to physically drawing these markings are numerous: Once inked, they cannot be removed. Certain games require different markings and some markings are irrelevant or distracting when playing other games. —When obligated to place a ball directly onto a marking, one's vision is naturally obstructed by the ball, so precise positioning is extremely challenging. —Cloth often stretches and shifts over time and cause the position of the markings to shift and/or fade away. [0017] Sticker markings often disrupt the path of the balls rolling over them. —It is not possible to easily configure. Lasers and projection patterns presented from directly above are useful to: —Precisely position a ball onto a designated spot. —Determine the winner of the lag. —Visualize lines and aiming points during practice—Organize training regimen and exercises—Can be reconfigured to adapt to whatever game or format is currently being played.

Players spend decades learning to understand exactly how to see and align themselves straight behind their shots before they can learn to shoot straight. It is a challenge and a struggle to understand the perspectives of each eye and position their bodies accordingly. Coupled together with the overhead mounted laser alignment line, a glass sphere, the size of a billiard ball, interacts with the laser beam overhead and provides an extremely clear and unique feedback to the player about precisely where their head is in relationship to the balls.

[0018] A glass (or acrylic) cue ball acts as a prism when interacting with the overhead light and refracts the light in a unique way to provide revealing alignment indications which are beneficial to help the practitioner understand the relationship of their head and eyes to the cue ball.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] Other objects and features of the present invention will become apparent from the following detailed description considered in connection with the accompanying drawings which disclose at least one embodiment of the present invention. It should be understood, however, that the drawings are designed for the purpose of illustration only and not as a definition of the limits of the invention.

[0020] In the drawings, wherein similar reference characters denote similar elements throughout the several views:

[0021] FIG. 1A is a plan view of a first embodiment;

[0022] FIG. 1B is a plan view of the embodiment of FIG. 1A;

[0023] FIG. 1C is a side cross-sectional view of the light;

[0024] FIG. 2 is a side view of the light positioned over a table;
[0025] FIG. 3 is a plan view of another embodiment of the light;
[0026] FIG. 4A is a top view of a table;
[0027] FIG. 4B is another top view of a table;
[0028] FIG. 4C is another view of a table;
[0029] FIG. 5 is another top view of a table;
[0030] FIG. 6 is a side perspective view of a table with a laser light shining on it;
[0031] FIG. 7 is a perspective view of a laser mic clamp for holding a laser light;
[0032] FIG. 8 is a perspective view of a spine clamp;
[0033] FIG. 9 is a perspective view of a foot spacer;
[0034] FIG. 10 is a perspective view of a laser sleeve;
[0035] FIG. 11 is a perspective view of a laser combined bracket;
[0036] FIG. 12 is a perspective view of a laser clamshell beefed up along the spine;
[0037] FIG. 13 is a perspective view of the cap and practice laser; and
[0038] FIG. 14 is a perspective view of the cap and practice laser magnet.

DETAILED DESCRIPTION

[0039] Referring to the drawings, FIG. 1A is a plan view of the first embodiment which includes a light frame **10** which includes a center extending arm **19** having a coupling **19.1** which is axially slidable on rails of the frame **10**. This arm **19** includes an extension section **19.2** which has inside a power and control feed **19.3**. There is a first light **19.4** which is an optional light which extends up above towards the ceiling which indicates that the device is “on”. In addition, there is a light **19.5** which is a laser light which shines down on ball maker **17.2** and which is used to either allow the placement of a triangle rack **21** or the marking of the cue ball on that spot.

[0040] The billiard table **14** includes a plurality of pockets **9.1, 9.2, 9.3, 9.4, 9.5,** and **9.6**. This table also includes another ball marker **17.1**. There are also a plurality of side ball placement markers on the rails of the table **15** including markers **15.1, 15.2, 15.3, 15.4, 15.5, 15.6, 15.7, 15.8, 15.9**.

[0041] FIG. 1B shows a plan view of the frame **10** which includes the arm **19**. In this case the pool table or billiard table **14** is shown in dashed dotted lines. The frame **10** includes sides **7, 11, 12** and **13**. The arm **19** obtains power from the power lines inside of the frame **10** such as shown in FIG. 1C which shows the cross-sectional view of these sections. For example, power line **19.3** obtains power from power supply **36** housed in frame **10**. It may be remotely switched via the power controller **36c** housed within the frame.

[0042] FIG. 1C shows a cross-sectional view of a frame which houses at least one LED light. The light frame **10** comprises a frame **20** which comprises a lid **26** with a groove for sliding clamps. There is a lip groove **22** for holding the lens **40** and filter which includes a corner connection to a lens **40**. There is also a segmented lid **26** to allow for containing various power drivers wiring and power distribution, computer controllers, and an exit hole **33** for RF antenna **34** to extend control of lighting (power, lumens, or color temperature) wirelessly in the remote range. A spacer bracket **29** comprising an aluminum heat sink **28** formed integral therein is positioned inside of the frame **20**. The heat sink can optionally have grooves or fins for adding surface area for aiding in cooling, wherein the spacer bracket is configured to support the LED circuit board **32** for housing the LED's **30**. The LEDs **30** are LED chips forming a light source. There is an AC/DC power supply driver **36** having an AC power cord **36a** which is configured to be plugged into a power supply (AC power supply). At the opposite end of the driver are low voltage DC power cables **35** to feed into the circuit board **32**.

One or more additional AC power cords **36b** tap from the external power line to provide power to any of the mounted peripherals (e.g., lasers, projects, etc.). Each powered peripheral device can be remotely switched via a wireless control unit **36c**. Full control of the LED lights as well as the laser and projectors onto the table surface can be coordinated and balanced wirelessly via remotely running control software.

[0043] The lens **40** is a downlight lens configured to focus the light provided by the LED's so that a focused beam is presented or produced from the frame **20**. In certain sections, there are light blocking filters **40b** which are configured to regulate the concentration of light on regions of the table, potentially applied to balance or compensate for external lighting conditions. For example, in this design, light blocking filters **40b** are positioned on the periphery of the lens.

[0044] FIG. **1C** also shows a transceiver **36c** disposed along line **36b** to control the output of line **36b**. In addition there is also shown transceiver **36d** disposed inside of power box **36** to control the power being fed to the LED's to control the brightness of the LED's. In addition, there is also a controller **136** such as a smartphone which is configured to send instructions to these transceivers **36c** and **36d**.

[0045] FIG. **2** is a side view of the light positioned over a table. In this view there is shown a cross-sectional view of frames **11** and **12** which produce the light for a table such as a billiard table. These frames **11** and **12** are positioned above the nose/bumpers of the table. The billiard table **14** comprises bumpers **16**. These bumpers **16** are wooden rails with a rubber cushion facing. Because a focused light is produced on these bumpers it creates a shadow from these bumpers thereby showing the presence of these bumpers more prominently because of the hard shadows that are produced by the lens **40**. In previous different designs, a frame would be coupled to a diffuser and not a lens. When a diffuser is used instead of a lens, the diffuser “diffuses” or softens and spreads the light rather than focusing the light thereby causing a softening of the contours of the bumpers which makes the bumpers less visible. Therefore, in order to accentuate the presence of the bumpers, a lens is used instead of a diffuser. This allows for the creation of shadow areas **16.1** and **16.2** underneath bumpers **16**. In addition, blocking filters such as blocking filters **40a** and **40b** are also used.

[0046] FIG. **3** is a plan view of another embodiment of the light which includes a rectangular frame **50**. The rectangular frame **50** is a light base platform having longitudinal extension **52** and latitudinal extension **51** for forming a substantially rectangular light. The frame has a lid **59** which is segmented for access to power distribution lines and RF antenna wires. There is a center beam **58** which is coupled at each end via clamps **53** and **63** to the latitudinal (shorter) extensions **51**. Coupled to this center beam **58** is a laser line projector **54** and a laser line **60**. This laser line projector **54** housed in a coupling **54.1** and is adjustable fore/and aft can sway have a fan angle. In at least one embodiment this laser line extends latitudinally. The laser line which is sent from the laser line projector **54** is for an extended line **54.2** extending across the length of the table and which is used for alignment of the light. The laser dot coupler **60.1** is for setting a laser light **61** for the rack point of the triangle. In essence these laser light sources are positioned over a center region of the table and are used for dispersing light to the table so that the outer frame **50** can be positioned above the table to create a substantially even distribution of light while still providing for substantial shadowing of the bumpers on the billiard or pool table. There are slidable mounts such as mount **65** which can be slidable along the latitudinal expanse **51** of the frame **50**. There is also a rib **66** which is coupled to the outer frame **50** via sliding bracket or mount **57** as well. The frame **50** is formed from the individual frames **7**, **11**, **12**, and **13** shown in FIG. **1B**. with each of the latitudinal expanse **51** and the longitudinal expanse **52** formed from these frames **20** (See FIG. **1C**) to create a substantially rectangular light frame housing LED lights which are configured to shine down on a table. This frame can be embedded into a ceiling such as a drop ceiling or hung from a ceiling as well. Rib **66** is coupled to center beam **58** via locking couplers **55** and **56**, with rib also being slidable on the longitudinal expanse **52** via sliding mount **57**. The locking couplers comprise a tubular spine coupling section (telescopic) with locking mechanism **56**. The center beam forms **58** forms a spine section with power/control which has wire holes for wire management. All of these wires are hidden inside the tubular spine for a clean and neat look. There is also a tubular spine segment protrusion **63.1** for coupling/locking extension arms at foot of the table. In addition, there are tubular spine segment protrusion for coupling/locking extension arms at head of table **65**

as well. Tubular rib segment **66** shows a protrusion for coupling/locking extension arms outside of table. In addition there is a laser line projector **69** coupled to coupling **61.1** and which is for projecting a laser line **14i**.

[0047] FIG. **4A** is a top view of a pocket billiards table showing the lighting effects that can be created on the table with the laser line projected by the laser line projector **54** (See FIG. **3**) on the table. For example, line **14i** is a laser projection line created by laser projector **69** positioned below laser **61**. This laser line **14i** is projecting onto the floor and ending on the table from laser line creator **54**. This line is used for footwork training. In addition, triangle **14a** is used as a three-segment set of lines using three-line drawing lasers (See laser **61** which is three separate line lasers forming a triangle **14a**) with cut off filters to control the length of the line laser. Line **14b** can be created by laser **54** is a laser line projection to form the head-string which is the line where the player is required to place the cue ball behind. This line is produced by an overhead laser line producer **61** using beam limiters or filters to cut off the beam at the edge of the bumpers.

[0048] FIG. **4B** is another top view of a pocket billiards table having extending arms **19** and **29**. These arms are coupled to the frame. Arm **29** is used to mount laser line producer **29.4** which is used to mark the head-string **14b**, or zone from where the opening break shot is taken. Arm **19** is used to mount laser line producer **19.6** and project a useful line for aligning the racking device **14.1** on the opening shot as well as re-spot balls during play. A laser line producer **19.7** is also positioned on arm **19** to create laser line **14i**.

Essentially these components are defined as follows: [0049] **14b**: Head-string laser projection produced by a single laser line producer **29.4**. [0050] **14h**: Rack alignment laser projection produced by a single laser line producer **19.6** spanning from the center marker of the end-rail to the foot-spot. This line acts as a guide to coincide with the centered markings **149a** and **149b** of a specially marked ball racking device **149** (e.g., triangle frame or template) to ensure that the arrangement of the racked balls at the start of each game is symmetrical to the table. [0051] **14i**: center line extending lengthwise along the center of the table created by the line laser **29.3**

[0052] There is also a laser **19.8** which creates a spot or marking (temporary) **14k** to mark the location of the placement of the cue ball on this end of the table.

[0053] FIG. **4C** is top view of a pocketless carom billiard table having a “spine” beam **58** extending arm **58** wherein the laser lines can show balk lines **14d** for special games. In this view there are multiple extending arms **29.1**, **29.2**, **29.6**, **29.7**. These arms each have laser line projectors **129.1**, **129.2**, **129.6**, and **129.7** respectively, for producing balk lines **14d** to mark playing zones within each corner of the carom billiards table.

[0054] On extending arm **58** the laser dot producer **141.1** can be used to show one or more spots such as the head spot(s) **14e**, (produced by a single laser dot) the center spot **14f**, as well as the foot spot **14g**. Other useful ball-marking spots **141** may be projected onto fixed positions on the table by mounting laser beam dot producers onto any number of tubular ribs **66** that are coupled to the spine and frame.

Essentially these components are defined as follows: [0055] **14d**: Balkline line projection produced by laser line producer **129.1**, **129.2**, **129.6**, and **129.7** directly mounted overhead (on arms with beam limiters). [0056] **14e**: Head spot laser projection produced by single laser beam dot **141.1** mounted onto the spine directly over the head-spot. [0057] **14f**: Center spot laser projection produced by single laser beam dot **141.2** mounted onto the spine directly over the center-spot. [0058] **14g**: Foot spot laser projection produced by single laser beam dot **141.3** mounted onto the spine directly over the foot-spot.

[0059] FIG. **5** is a top view of a table **14** showing projection images or artwork **14L** on the floor outside of the boundaries of the table as well as generated images or artwork projected within the boundaries of the table. The projection device **39a** is coupled to arm **38a**. Arm **38a** is coupled to a frame such as frame **20**. This view shows extending arm **38** and **38a** extending out from frame. Arm **38** has a projector **39** which shines a logo down on a floor area adjacent to the billiard table

14. Arm **38a** has a projector **39a** which projects generated images or artwork **14M** directly onto the billiard playing surface.

[0060] FIG. **6** is another top view of a table **14** which shows a frame **20** which has an arm **19** coupled thereto. There is a laser line creator **19.7** which creates a laser line **14i** along the longitudinal axis of the table. There is a table surface shown **142** as well. Line **14i** can be used for foot placement when a user is practicing their shots. There are thus shown two feet placements **159** and **161** shown wherein foot placement **161** is shown along line **14i**.

[0061] FIG. **7** is a perspective view of a laser clamp for holding a laser light such as laser lights **54**, **60** and **61**. This clamp **200** includes a body section **204** which is substantially concave and has a hole **201** for having a locking pin secure to a laser shaft which is housed inside of the body of the clamp **200**. The head section of this clamp includes two openings **202** and **203** which allow this laser clamp to be adjusted laterally as well as rotationally.

[0062] FIG. **8** is a spine clamp for clamping both the beam **58** and the rib **66**. FIG. **8** is a perspective view of a spine clamp **210**. This spine clamp is essentially similar to the clamps or mount **53** and **57** which clamp the beam **58** and rib **66** to the remainder of the frame **50**. This clamp **210** includes a body section **214**, a tubular or cylindrical head section **212** as well as a slot **215**. Slot **215** allows this clamp to be slid axially along the frame **50** to allow for lateral or longitudinal adjustment along a track disposed on the back side of the frame **50** opposite the side which contains the lens **40**. The following are definitions of the components shown in FIG. **8**: **210**: Spine-mount clamp device; **212**: Flexible sleeve to tighten around tubular spine segments; **214**: Embedded tightening bolt to tighten clamp onto spine tightly, **215**: Frame-bracket clasp to lock spine-mount down onto frame (using spine-mount foot-spacer lock)

[0063] FIG. **9** is a perspective view of a foot spacer **220** having a bridge section **224** and an open section **222**. Foot-spacer **220** slides under the spine-mount clamp between clamp and frame to lock spine-mount into place **222**: Handle-space **224** is configured to pull foot-spacer away from spine-mount to release locking action: Handle to push foot-spacer tightly under spine-mount to lock mount into place on frame. The following are the definitions of the components shown in FIG. **9**: **220**: There is a foot-spacer **222** which slides under Spine-mount clamp between clamp and frame to lock spine-mount into place: Handle-space to pull foot-spacer away from spine-mount to release locking action. **224**: Handle to push foot-spacer tightly under spine-mount to lock mount into place on frame.

[0064] FIG. **10** is a perspective view of a laser sleeve **230** which includes a block section **234** a hollow section **236** and a loop section **232**. The block section **234** having the hollow section **236** is configured to house a cylindrical shaped laser beam device such as laser beam devices **54**, **60** and **61**. Loop section **232** is configured to receive a lock which clamps and locks the block section **234** to a laser. The following are a list of components which are shown in FIG. **10**: **230**: Adjustable angle laser-line device holster to attach to spine; **232**: Tightening screw/nut to clamp onto laser device; **234**: Side of laser holster with 3 hole(s) for fine-tuning screws used to calibrate direction of laser **236**: Inner chamber of laser holster.

[0065] FIG. **11** is a perspective view of a laser combined bracket **240** which includes a body section **242**, a loop section **249** allowing the laser to be rotated about a shaft disposed inside of the loop section. There is also an opening **244** in a block section **246** wherein block section **246** includes a lock opening **248** for receiving a lock to clamp the block **246** around a laser housed inside of opening **244**. The following are the components shown in FIG. **11**: **240**: Fixed angle alignment laser beam holster; **242**: Side of laser holster with 3 hole(s) for fine-tuning screws used to calibrate direction of laser; **244**: Slides onto spine for repositioning fore/aft. **246**: Spine mounting clamp; **248**: Spine clamp inset screw; **249**: Tightening screw/nut to clamp onto laser device

[0066] FIG. **12** is a perspective view of a laser spine sliding mount **250** disposed along the spine. This laser spine mount includes a locking body section **254**, an opposite lock **256** to attach an adjustable-angle laser holster as well as a central column **252** for tightening the mount to the spine.

The following are components shown in FIG. 12: **250**: Adjustable sliding spine mount; **252**: Slides onto spine for repositioning fore/aft; **254**: Locking body section; **256**: Locking adjustable angle device attachment.

[0067] FIG. 13 is a perspective view of the rotatable practice laser device mounting portion **260** designed to attach to the practice laser spine clamp **270** from FIG. 14. This practice laser connector **260** has a cavity **261** designed to carry donut-shaped magnets, a cylindrical section **263** that fits snugly but rotates freely around **278** (from FIG. 14), a device mount tab **264**, and a hole **262** for a screw to hold magnets into place. The following are components shown in FIG. 13: **260**: Rotatable magnetic practice laser attachment device; **261**: Cavity to hold donut shaped magnets; **262**: Screw to hold magnets into place; **263**: Wall to fit cap over cylinder protrusion **278** (from FIG. 14); **264**: Tab to mount laser holster device; **265**: Hole to adjust and hold laser mount holster at set angle.

[0068] FIG. 14 is a perspective view of the spine clamp portion **270** of the rotatable practice laser device mount. This device includes a body **272** having an opening **274** for mounting onto the spine. There is a locking hole **276** for receiving a screw to tighten the clamp **274** around the spine. The cylindrical protrusion of the device **278** carries steel washers that attract to the magnets contained in **261** (FIG. 13), thus holding the laser mount in place while allowing cylindrical section **263** to rotate freely round it. There is also an opening **279** to accommodate the magnet mounting screws from **260**. The following are components listed in FIG. 14: **270**: Spine clamp for rotatable practice laser attachment device; **272**: Locking block with nut; **276**: Locking screw; **274**: Opening for spine; **278**: Cylindrical protrusion with steel washers inside; **279**: Opening to accommodate screws that hold **262** magnets into place.

[0069] Thus, there is created a design for a lighting system for a billiard table which is configured to allow a user to adjust and align the lighting system over a billiard table to provide substantially symmetrical light by using laser guides such as a laser line or a laser dot.

[0070] Accordingly, while at least one embodiment of the present invention have been shown and described, it is to be understood that many changes and modifications may be made thereunto without departing from the spirit and scope of the invention as defined in the appended claims.

Claims

1. A lighting system for a billiard table comprising: a frame; a plurality of lights disposed in the frame; at least one lens coupled to the frame, said at least one lens for focusing a light produced by at least one light disposed in the frame.
2. The lighting system as in claim 1, wherein said plurality of lights comprise at least one LED light.
3. The lighting system as in claim 1, wherein said frame is a substantially rectangular frame.
4. The lighting system as in claim 3, wherein said frame comprises at least two longitudinally extending sections which are spaced substantially parallel to each other and at least two latitudinally extending sections extending substantially transverse to the longitudinally extending sections wherein said latitudinally extending sections are coupled to the longitudinally extending sections.
5. The lighting system as in claim 4, further comprising a center beam extending across a center region of the frame between at least one of the at least two longitudinally extending sections.
6. The lighting system as in claim 5 further comprising at least one rib extending transverse to the center beam.
7. The lighting system as in claim 5, further comprising at least one additional light coupled to the center beam.
8. The lighting system as in claim 7, wherein said at least one additional light is a laser light.
9. The lighting system as in claim 8, wherein said at least one additional light is a laser light configured to produce a laser line.

10. The lighting system as in claim 8, wherein said at least one additional light is a laser light configured to produce a dot.
